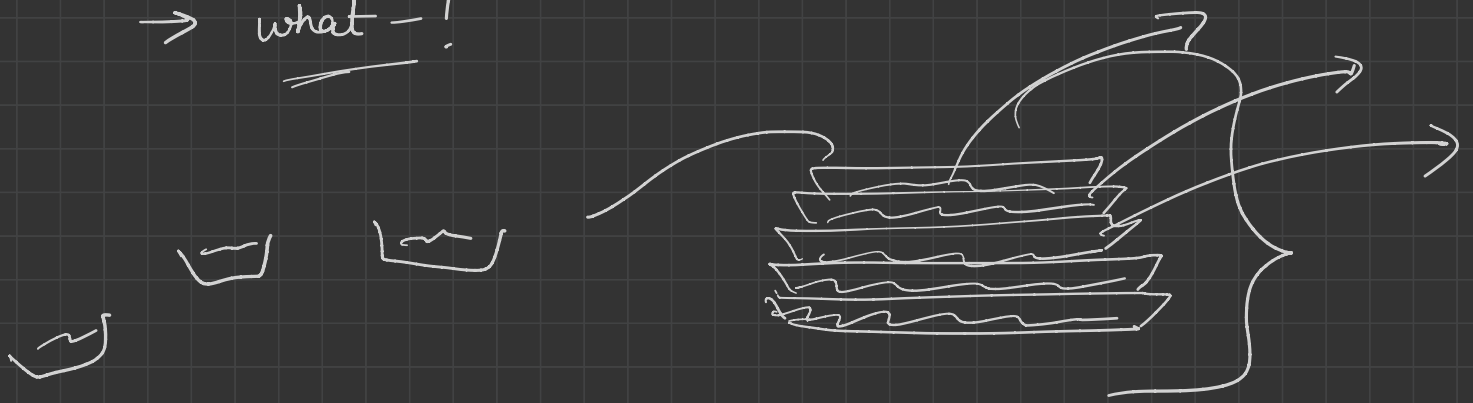
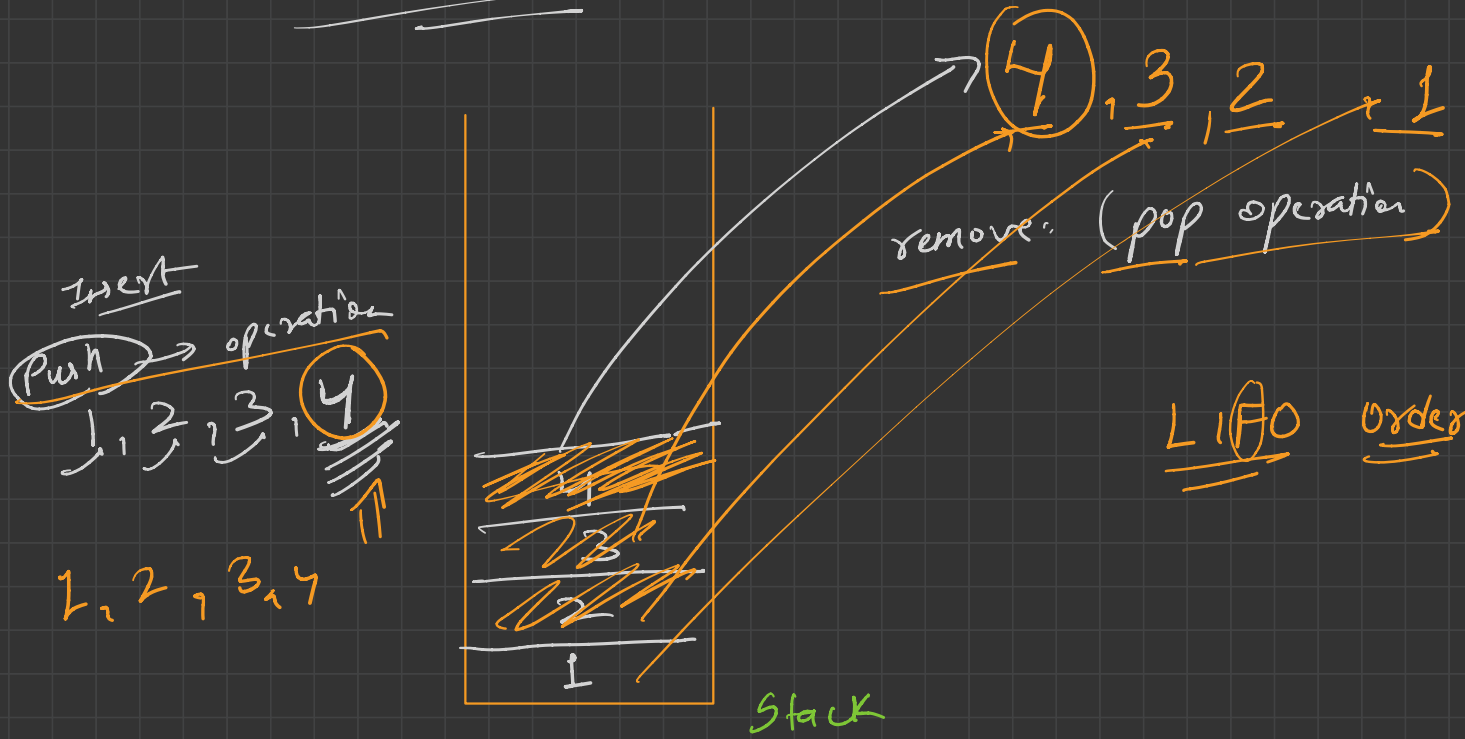
Stack

→ what -!



→ LIFO → Last In first Out

Stack → D.S



Stack:- D.S → LIFO order



Operation:-

- push
- pop
- peek → top element
- isEmpty empty



STL

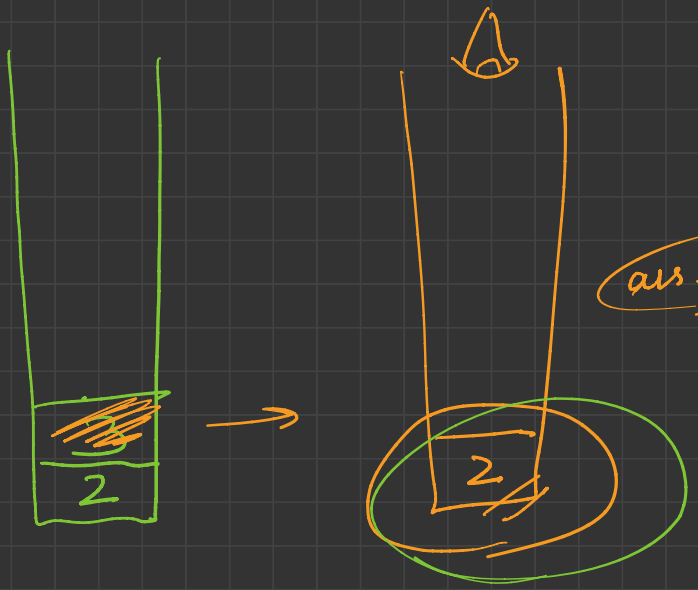
creation → `stack <int> s;`

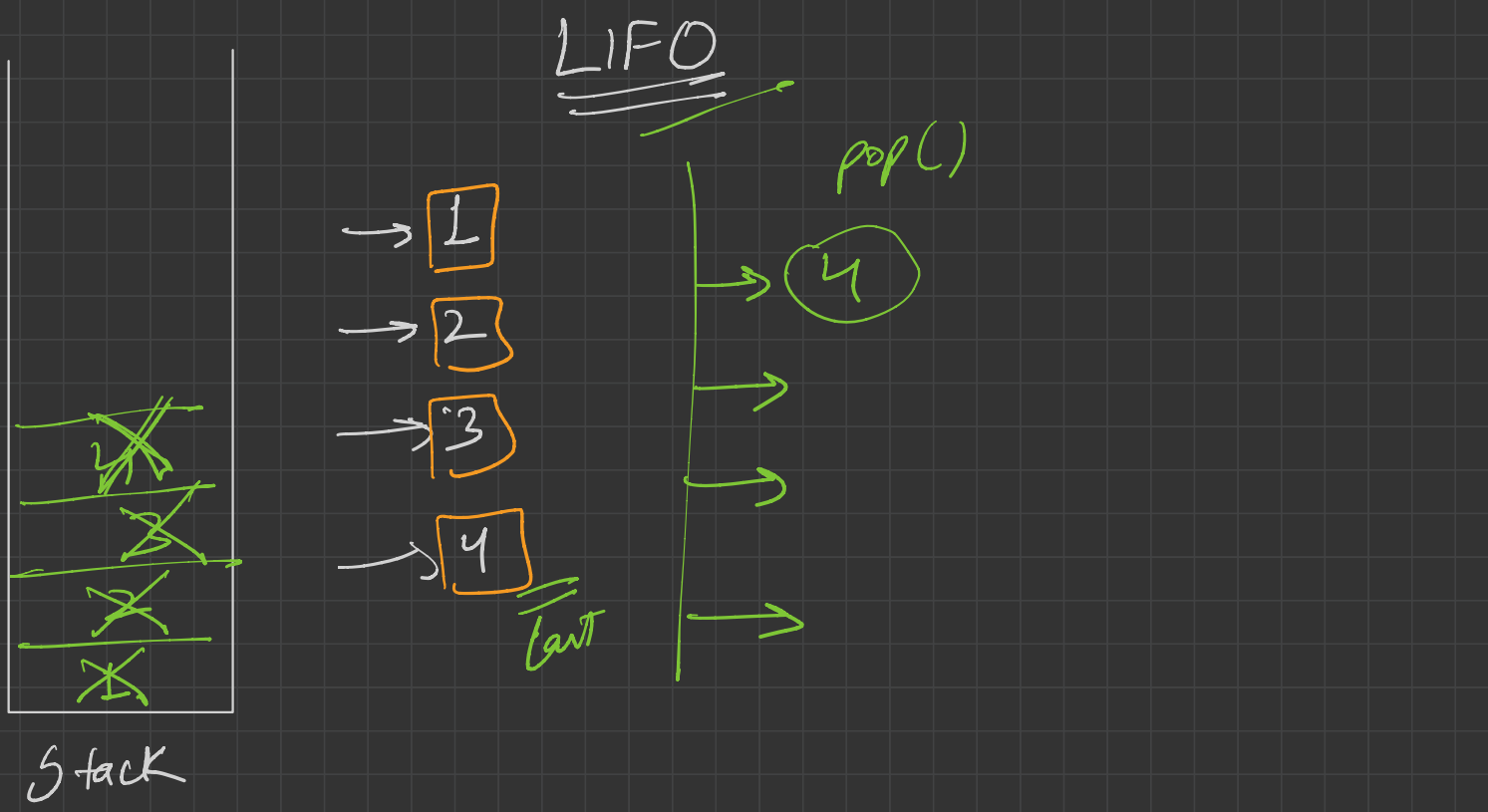
push → `s.push(2);`

pop → `s.pop();`

Peek/top → s.top() → top

is empty → s.empty() → T/F




LIFO
pop()
1
2
3
4
last
Stack

→ what → ?

→ Stack src → code

STL α

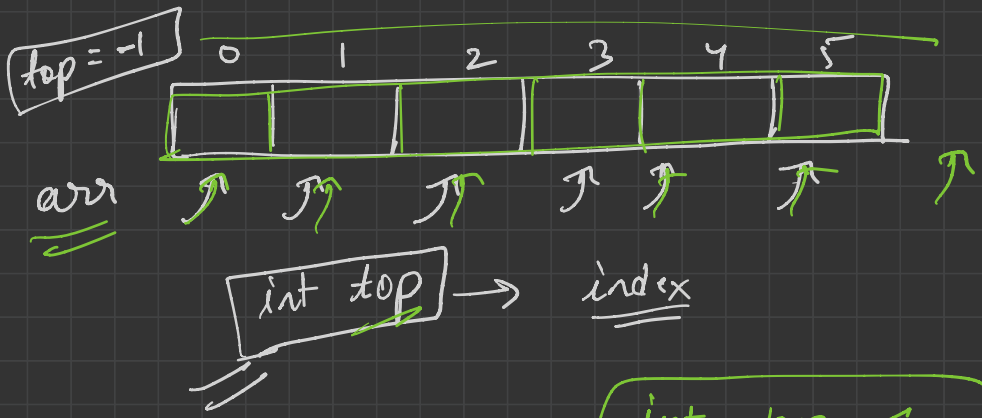
→ Stack Implementation

class stack → =

Stack → Impl

→ array =

→ Linked List → Homework =


top = -1
0
1
2
3
4
5
arr
int top → index
int top
int *arr
int size
push
push operation
top = -1
top = 0
top
push (22)

↳ check → is space available

T → Insert

F → Stack Overflow

Insert

top ++ ;

arr [top] = 22;

top = -1 → stack is empty

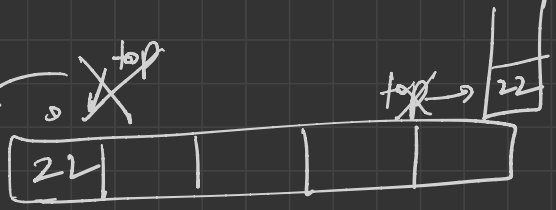
pop()

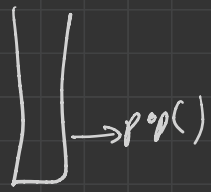
↳ top >= 0

check element is present

T → top --

F → Stack Underflow

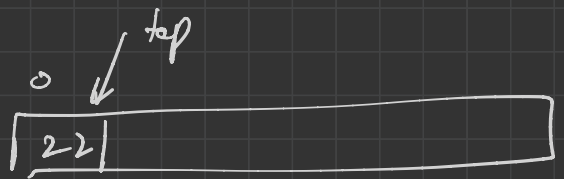




empty()

top == -1 → stack empty

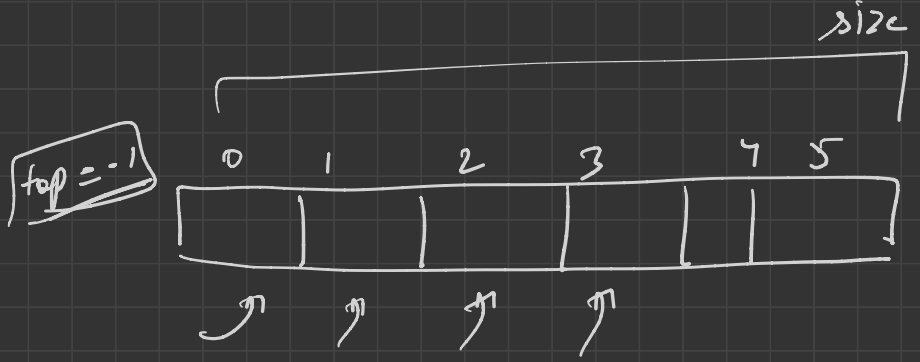
top()



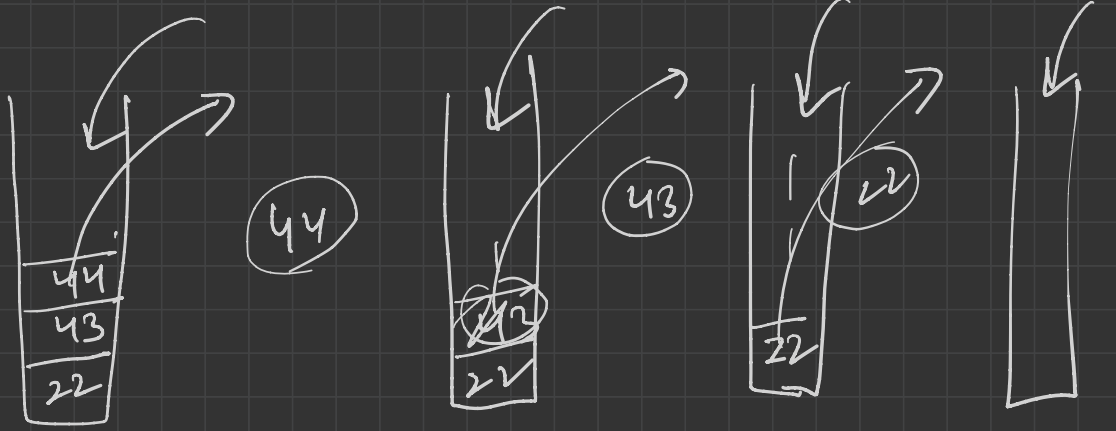
return arr[top];

top = -1

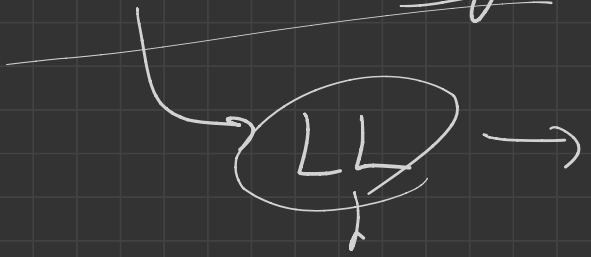
top < size



size - top > 1 → atleast 1 empty space



→ Stack Impl → array



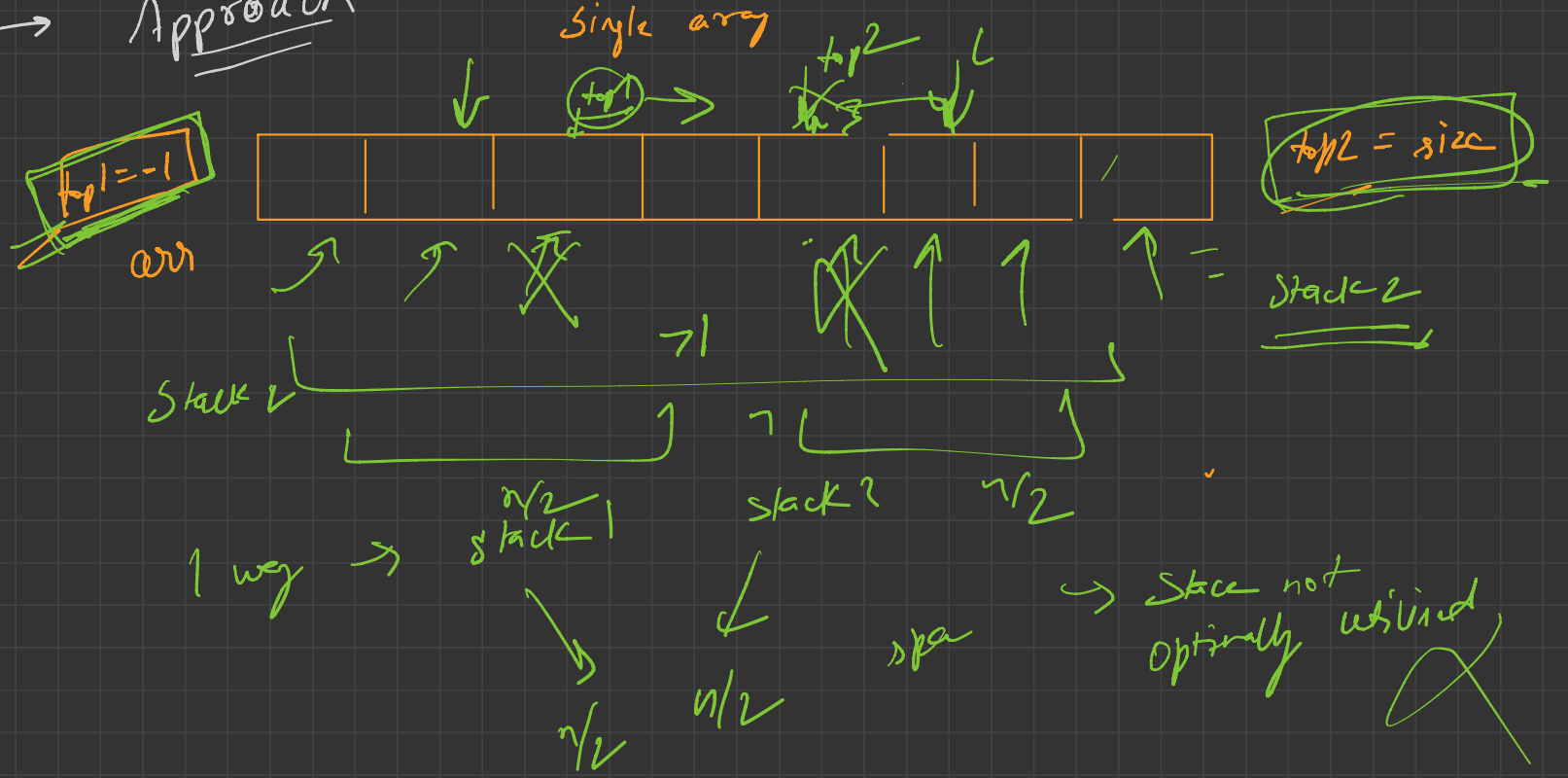
→ Homework

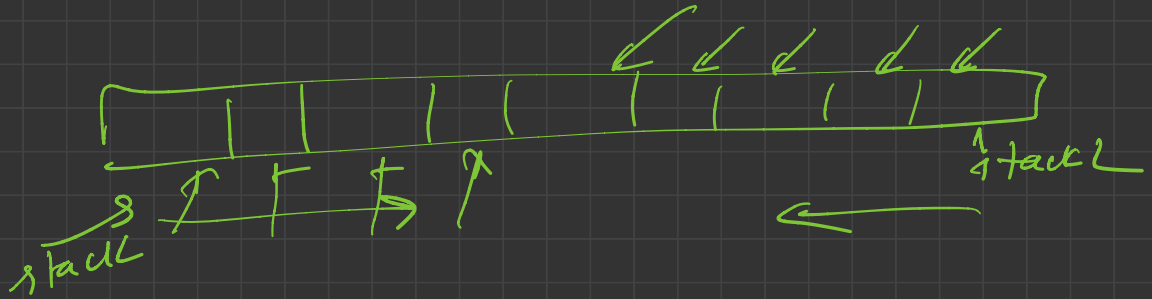
Easy

→ Easy Question

↳ 2 Stacks in an array

→ Approach


Single array
top1
top2
L
top1 = -1
top2 = size
arr
Stack 2
Stack 1
n/2
n/2
n/2
n/2
1 way →
Stack 1
Stack 2
space
→ Space not optimally utilised


stack
stack